

5. BSc CSIT Curriculum (Semester-wise)

First Semester

Introduction to Information Technology, C Programming, Mathematics I, Physics, Digital Logic

Second Semester

Discrete Structures, Object Oriented Programming, Microprocessor, Mathematics II, Statistics I

Third Semester

Data Structures and Algorithms, Numerical Method, Computer Architecture, Computer Graphics, Statistics II

Fourth Semester

Theory of Computation, Computer Networks, Operating Systems, Database Management System, Artificial

Intelligence

Fifth Semester

Design and Analysis of Algorithms, System Analysis and Design, Web Technology, plus Elective I (e.g.

Cryptography, Simulation and Modeling, Multimedia Computing, Wireless Networking, Image Processing,

Knowledge Management, Society and Ethics in IT, Microprocessor Based Design)

Sixth Semester

Software Engineering, Compiler Design and Construction, Net Centric Computing, plus Elective II (e.g.

E-Governance, Technical Writing, Applied Logic, E-Commerce, Automation and Robotics, Neural Networks,

Computer Hardware Design, Cognitive Science)

Seventh Semester

Advanced Java Programming, Data Warehousing and Data Mining, Principles of Management, Project Work,

plus Elective (e.g. Information Retrieval, Database Administration, Software Project Management, Network

Security, Digital System Design, International Marketing)

Eighth Semester

Advanced Database, Internship, plus Electives (e.g. Advanced Networking with IPv6, Distributed Networking,

Game Technology, Distributed and Object-Oriented Database, Cloud Computing, GIS, Decision Support

Systems, Mobile Application Development, Real-Time Systems, Network and System Administration, Embedded

Systems Programming, International Business Management)

Note: Full syllabus PDFs for each subject and the official Prospectus are downloadable from the official BSc CSIT program page

(academiacollege.edu.np/bscsit).